DIVYANSHU GOYAL

Applied Scientist | Multimodal LLM Training and Evaluation
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SUMMARY

I am an Applied Scientist at Adobe, and my interests lie in multimodal language models. At Adobe, I train and evaluate them for creative work. The work often hinges on subjective quality, like whether a photo edit looks better or marketing copy feels original, so I spend a lot of my time building evaluations that can be used to reliably train and compare models. I have shipped this work in Adobe GenStudio and published it at CVPR, ICML, and AAAI workshops.

EXPERIENCE

Applied Scientist – Adobe Inc., San Jose, CA June 2021 – Present

Vision-Language Model Training & Evaluation

- Led **ATune** (under review): a 54M-parameter model that learns photo retouching from 10M unedited photographs and operates off-the-shelf editors by setting their sliders. With 157 raters, its edits beat GPT-6 Astra using the same editor (**54.3%** of 3,524 votes, 95% CI 50.3–58.3) and tied with Claude Opus 5.
- Fine-tuned **Gemma-3-12B** with LoRA to score marketing-image quality, training on 8× H100 (DDP, BF16, FlashAttention-2, Liger kernels).
- Grounded VLM quality reasoning in 14+ pixel-level metrics (sharpness, exposure, color balance); the resulting interpretable scorer reaches **MAE 0.77** (Pearson r ≈ 0.88) vs. 1.34 for zero-shot GPT-5.4 and 1.95 for Claude Opus 4.7. Also built **MQCore**, an evaluation score for marketing-image quality.

Evaluation Research & Benchmarking

- Created **DistortBench** (first author; CVPR 2026 and ICML 2026 Workshops): a 13.5K-question benchmark over 27 distortion types and 5 severity levels. Evaluated 18 VLMs with bootstrap confidence intervals; the best model (Qwen3.5-27B) scores **61.9%** (chance 25%), above the average annotator (60.2%) and just below a 3-annotator majority vote (65.7%).
- Co-authored **JudgeBreak** (ICML 2026 Workshop): across 95 subjective settings, linear probes on Qwen3-32B hidden states with cross-validated layer selection reach **78.1%** mean pairwise accuracy vs. 70.1% for the best reward model or zero-shot GPT-5.4 judge.
- Co-authored a curiosity-driven **LLM-as-a-judge** (AAAI 2026 Workshop, Oral) that personalizes creative-writing evaluation by modeling the judge's surprise at each expert's explanation, raising mean F1 from 0.37 to 0.62 and Pearson r from 0.15 to 0.56 over an SFT baseline. Trained its baseline models.

Production Multimodal Systems & Agents

- Prototyped **Adobe GenStudio for Performance Marketing** (launched October 2024, now serving Fortune 500 clients including Microsoft and AT&T), then designed its multimodal brand service, now deployed in 3 regions, for VLM content validation, brand-DNA extraction, and retrieval.
- Built a **multimodal Photoshop agent** that edits images from natural-language instructions, running an observe-think-act loop over 49 tools with vision grounding and a quality-critic refinement step.

Machine Learning Engineer – Adobe Inc., Bangalore, India July 2016 – August 2019

- Co-developed Adobe's production **ML prediction service**, which sustains 3,500 QPS/node at 0.3–0.6 ms app-side p99 latency (Adobe Tech Blog).
- Built an end-to-end model lifecycle platform: versioned deployment, live monitoring, and phased rollout across dev, staging, and production.

SELECTED PROJECTS

VibeNanoChat: GPT-2 Scale LLM Pretraining from Scratch (code) 2026
Decoder-only transformer (RoPE, RMSNorm, sliding-window attention) with one-knob depth scaling and a custom distributed **Muon** optimizer (ZeRO-2 state sharding), trained in BF16 with FlashAttention-3 on 8× H100 at over 1M tokens/sec. Fit isoFLOP curves across 40+ runs (44M to 522M parameters) to find compute-optimal model sizes.
Write-ups: Going Fast and A Lazy Engineer's Guide to Scaling Transformers.

PUBLICATIONS

DistortBench: Benchmarking Vision Language Models on Image Distortion Identification
D. Goyal, A. Eppa, V. Bannihatti Kumar CVPR 2026 and ICML 2026 Workshops | arXiv:2604.19966

JudgeBreak: Is Taste Hidden in the Layers? Stress-Testing Reward Models and LLM Judges with Representation-Based Subjective Evaluation
V. Bannihatti Kumar, A. Narang, M. Anand, A. Eppa, **D. Goyal** ICML 2026 Workshop | OpenReview

Curiosity-Driven LLM-as-a-judge for Personalized Creative Judgment
V. Bannihatti Kumar, **D. Goyal**, A. Eppa, N. Bhandari AAAI 2026 Workshop, Oral | arXiv:2510.05135

PATENTS

Guiding Language Translation with Translation Documents Using ML
D. Goyal, A. Eppa, M. Anand 2026 | US20260050753
Splits translation guidelines into language-agnostic facets and scores how well translations follow each one.

Brand-Aligned Marketing Content Generation
D. Goyal, M. Anand, J. Mathew 2025 | US20250117829
Turns unstructured brand material into structured brand data for on-brand content generation.

Cultural Adaptation of Images Using Generative AI
A. Eppa, M. Anand, **D. Goyal** 2026 | US20260051084
A VLM describes an image, the text is adapted to a target culture, and a text-to-image model renders it.

Parameter-Efficient Fine-Tuned Models
Y. Wang, N. Vangala, M. Anand, K. K. Jain, J. Mathew, E. Jose, **D. Goyal**, et al. 2025 | US20250217193
Serves many fine-tuned models by loading small adapter layers into preloaded base models on demand.

TECHNICAL SKILLS

Research: VLM Training & Evaluation, LLM Pretraining & Scaling Laws, Benchmark Design, Human Preference Studies, LLM-as-a-Judge & Reward Models, Linear Probing, Bootstrap Statistics

Tools: Python, PyTorch, Distributed Training (DDP, ZeRO-2), torch.compile, FlashAttention-2/3, Liger Kernels, Muon, Mixed Precision (BF16, TF32), LoRA, Hugging Face Transformers/PEFT, Weights & Biases

EDUCATION

Georgia Institute of Technology – M.S. Computer Science (Machine Learning), GPA 4.0 (2019 – 2021)

BITS Pilani – B.E. Computer Science, M.Sc. Mathematics, GPA 8.75/10 (2011 – 2016)